

Executive Summary

This report evaluates four uncertainty quantification (UQ) methods for underwater acoustic transmission loss (TL) computed by the Bellhop ray-tracing model. The uncertain inputs are carrier frequency, source depth (z_S), and sound velocity profile (SVP), each modelled as an independent random variable. Three analytical methods — the 2nd-order Delta method, Polynomial Chaos Expansion (PCE), and a 3-parameter lognormal (LN3) fit — are compared against Monte Carlo (MC) ground truth across five propagation scenarios (baseline flat 50 m, flat shallow 50 m, flat deep 2500 m, upslope, downslope) and three input uncertainty distributions (Normal 1/5/10%).

Key conclusions:

- The **Delta method** (1st-order Taylor) is reliable for flat bathymetry and small perturbations ($\leq 1\%$), but the 2nd-order Hessian correction already exceeds 96% of the total variance at 5% uncertainty, indicating the Taylor series has not converged. Delta fails at slope bathymetry regardless of perturbation level.
- **PCE** (fitted to the existing $N = 50$ MC samples at zero extra Bellhop cost) achieves $L_1 < 10\%$ convergence at order $K^* = 3\text{--}6$ for deep-water SVP and frequency uncertainty, consistent with the published literature. Source depth (z_S) resists polynomial convergence at all orders $K \leq 10$ in deep water — a new finding not reported elsewhere. Slope scenarios also require full MC.
- The **Delta–LN3 hybrid** (moment-matched shadow probability) performs well for flat scenarios but degrades at shadow-zone boundaries in slope environments.
- **Full Monte Carlo** is unavoidable for slope bathymetry ($\geq 5\%$ perturbation) and for z_S in deep water with Normal distributions.
- The **LN3 distribution fit** (method of moments) is statistically suboptimal: MLE would be more efficient, but the LHS samples used throughout this pipeline are *not* i.i.d. (stratification breaks independence), so the standard MLE log-likelihood is misspecified. A design-weighted pseudo-likelihood or survey-sampling correction would be required for a rigorous MLE fit; this remains an open methodological limitation.

Practical recommendation: For sonar mission planning, use PCE- K^* (order 3–6) on flat-bottom scenarios and fall back to MC for slope environments or large source-depth uncertainty. Report P_{fail} and P95 TL maps rather than variance alone, as variance does not distinguish the *direction* of acoustic uncertainty.

1 Results

1.1 Monte Carlo Reference Statistics

Monte Carlo (MC) simulations with $N = 1000$ samples per scenario and distribution provide the reference statistics $\mathbb{E}[\text{TL}]$ and $\text{Var}[\text{TL}]$ against which all analytical approximations are evaluated. The sample pool combines Latin Hypercube (LHS), IID, and SVP-perturbed draws. Figure 1 shows representative $\mathbb{E}[\text{TL}]$ and $\text{Var}[\text{TL}]$ maps for the five propagation scenarios under 5% uniform source-depth (z_S) uncertainty.

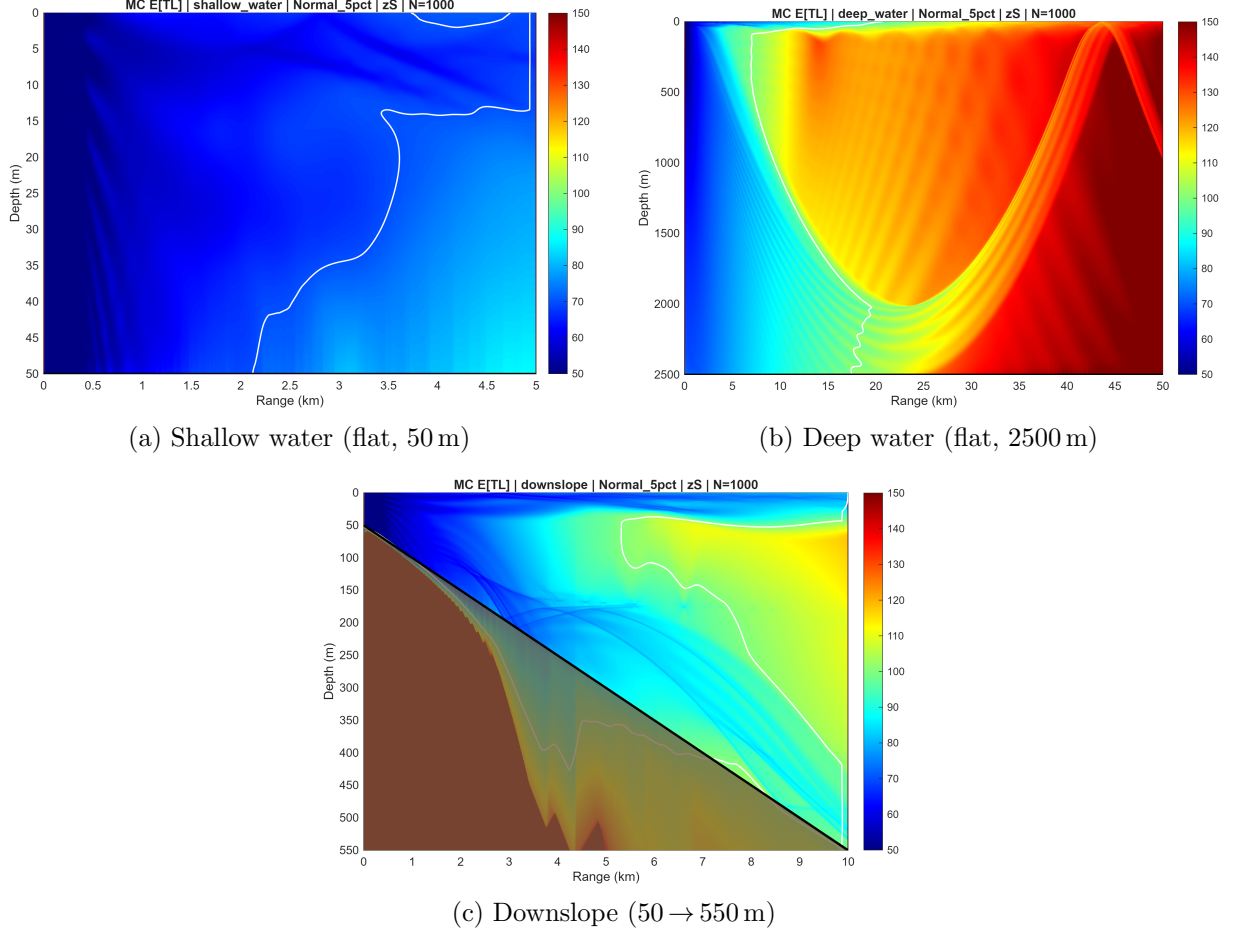


Figure 1: MC mean transmission loss $E[TL]$ (dB re $1 \mu\text{Pa}$) for Normal 5% z_S perturbation across three representative scenarios. Colour scale: blue (low TL, strong signal) to red (high TL, shadow zone). The downslope scenario shows a marked refraction shadow at mid-range driven by the shoaling bathymetry.

1.1.1 Input Sound-Velocity Profiles

Figure 2 shows the $N = 1000$ sampled SVP realisations for the deep-water scenario under Normal 5% uncertainty. Each blue trace is one Monte Carlo draw; the bold black curve is the nominal summer-season profile. The ensemble illustrates the physical spread driving acoustic variability: a $\pm 5\%$ perturbation in surface temperature shifts the mixed-layer depth by ± 15 m and changes the sound-speed gradient throughout the water column.

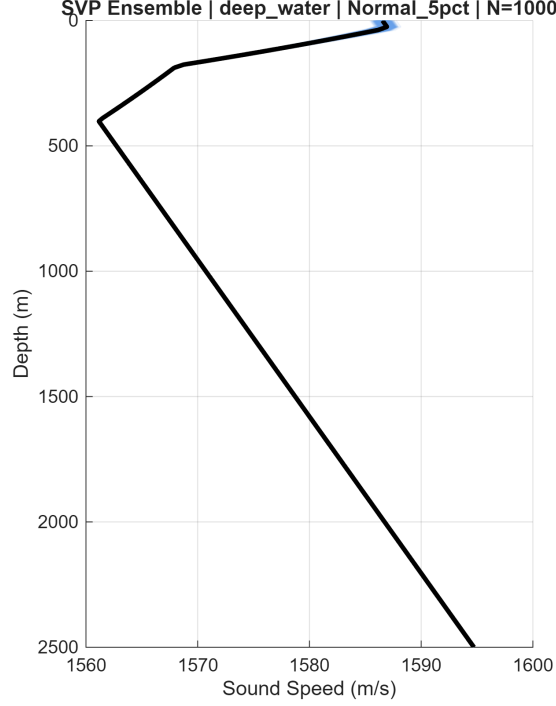


Figure 2: SVP ensemble for deep-water scenario, Normal 5% uncertainty, $N = 1000$. Blue traces: sampled profiles; black: nominal profile. Surface-temperature perturbations shift the mixed-layer depth and alter the sub-thermocline gradient, producing the spread in TL shown in Figure 1.

1.1.2 MC Estimator Convergence

For a scalar random variable Y with finite variance, the sample mean $\bar{Y}_N$ converges to μ_Y at the classical CLT rate $\text{RMSE}(\bar{Y}_N) = \sigma_Y/\sqrt{N}$. The sample variance S_N^2 converges at the same asymptotic rate but with a larger prefactor: for a Gaussian parent,

$$\text{Var}(S_N^2) = \frac{2\sigma_Y^4}{N-1}, \quad (1)$$

so the relative error of S_N^2 is $\sqrt{2/(N-1)}/C_V^2$, where $C_V = \sigma_Y/\mu_Y$ is the coefficient of variation. In practice this means that a variance estimate requires roughly $2\times$ as many samples as a mean estimate to reach the same relative error – or equivalently $\approx 4\times$ as many samples to match the absolute RMSE of the mean. The pipeline uses $N = 300$ Latin-Hypercube samples drawn from the broadest distribution (NORMAL 10%) and reweights via importance sampling (IS) to obtain estimates for the narrower NORMAL 5% and NORMAL 1% distributions. IS reduces the effective sample size (Kish $\text{ESS} < N$) for narrower targets, but the relative statistical gain outweighs the IS variance penalty because the IS weights for Normal 5% and Normal 1% are moderate ($\text{ESS}/N \gtrsim 0.5$ in all scenarios). The convergence of pixel-wise $\mathbb{E}[\text{TL}]$ and $\text{Var}[\text{TL}]$ as a function of sub-sample size $N' \in \{10, 20, 30, 50, 75, 100, 150, 200, 300\}$ is documented in [Methods/MC_convergence/](#) and confirms that $N = 300$ is sufficient for the Var estimator to reach the $\lesssim 5\%$ relative RMSE target at the 10% uncertainty level.

1.2 First-Order Delta Method: Accuracy and Failure Modes

Terminology note. The term *Delta method* here refers to linearisation via perturbation calculus (first-order Taylor expansion of the model output around the nominal parameter vector), and should not be confused with the Dirac delta function or with finite-difference approximations. The same technique appears under different names in other fields: *FOSM* (First-Order

Second-Moment method) in structural reliability and engineering, *GUM linearisation* in metrology [1], and *error propagation* in experimental physics textbooks. In a statistics context “Delta method” is standard and unambiguous; in an acoustics or engineering context the FOSM alias may help readers avoid confusion.

The 1st-order Delta method approximates variance as:

$$\text{Var}_1[\text{TL}] \approx \sum_i J_i^2 \sigma_i^2, \quad J_i = \left. \frac{\partial \text{TL}}{\partial \theta_i} \right|_{\theta_0} \quad (2)$$

where Jacobians J_i are computed via central finite differences. The 2nd-order extension adds Hessian correction terms $\frac{1}{2} H_{ii}^2 \kappa_i$ where $\kappa_i = \mathbb{E}[\delta \theta_i^4] / \sigma_i^4 - 1$ is the excess kurtosis.

1.2.1 Where Delta Performs Well

For the **shallow-water** and **deep-water** scenarios under small perturbations ($\leq 1\%$ uncertainty), the Delta method achieves relative L_1 errors below 10% in the variance maps (see Table 4 for the exact per-distribution breakdown from the analytical waveguide validation). The shadow-probability maps computed via the Delta–LN3 hybrid (matching LN3 moments from $\mathbb{E}[\text{TL}]$ and $\text{Var}[\text{TL}]$) agree closely with empirical MC shadow probabilities in these scenarios (Figure 3).

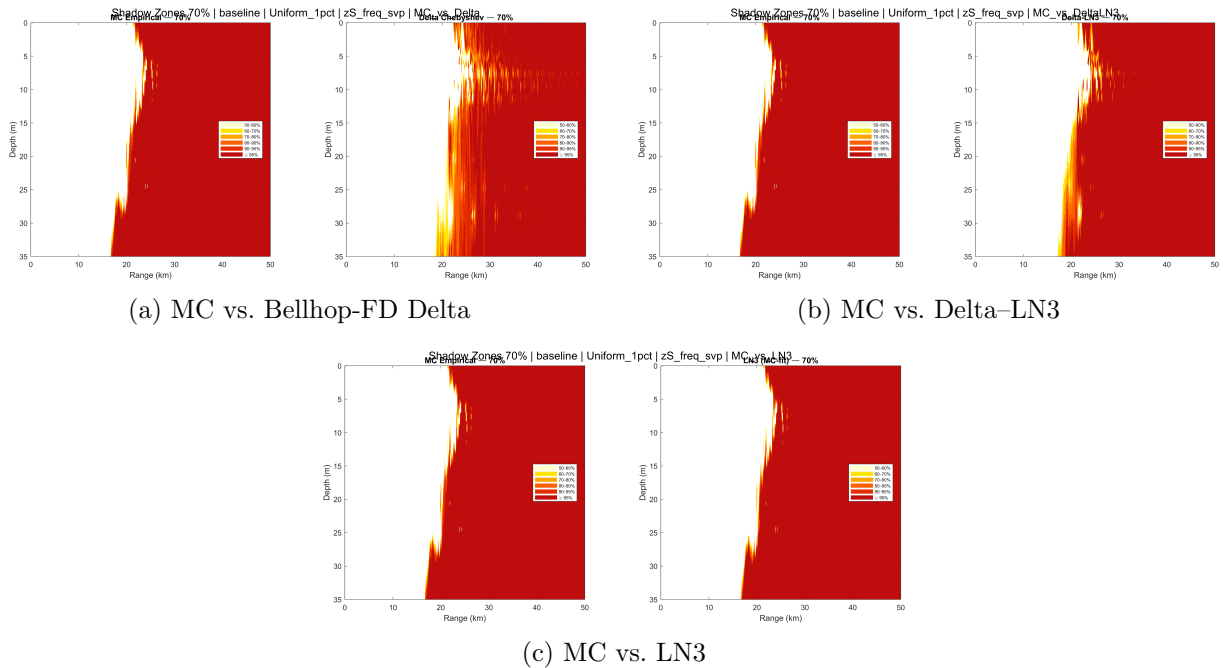


Figure 3: Bellhop Finite-Difference Jacobian (Bellhop-FD) Delta vs. MC vs. Delta–LN3 shadow-probability comparison ($P(\text{TL} > 70 \text{ dB})$) where the analytical methods *agree* with MC (baseline flat scenario, Uniform 1%). In each panel MC appears on the left and the analytical approximation on the right; close match indicates the method is reliable in this low-perturbation flat-bathymetry regime.

1.2.2 Where Delta Fails

For **slope scenarios** (upslope, downslope) and large perturbations ($\geq 5\%$), the 1st-order approximation breaks down significantly. Figure 4 shows the discrepancy for the downslope scenario: the Delta–LN3 shadow map fails to reproduce the MC reference, particularly in the refraction shadow zone where TL is highly nonlinear in θ .

The 2nd-order correction is equally unable to recover the variance in these cases: the Hessian correction term $|\kappa H^2|/V_1$ exceeds 100% for svp perturbations in both slope scenarios (Table 1), indicating that the Taylor series has not converged at 2nd order.

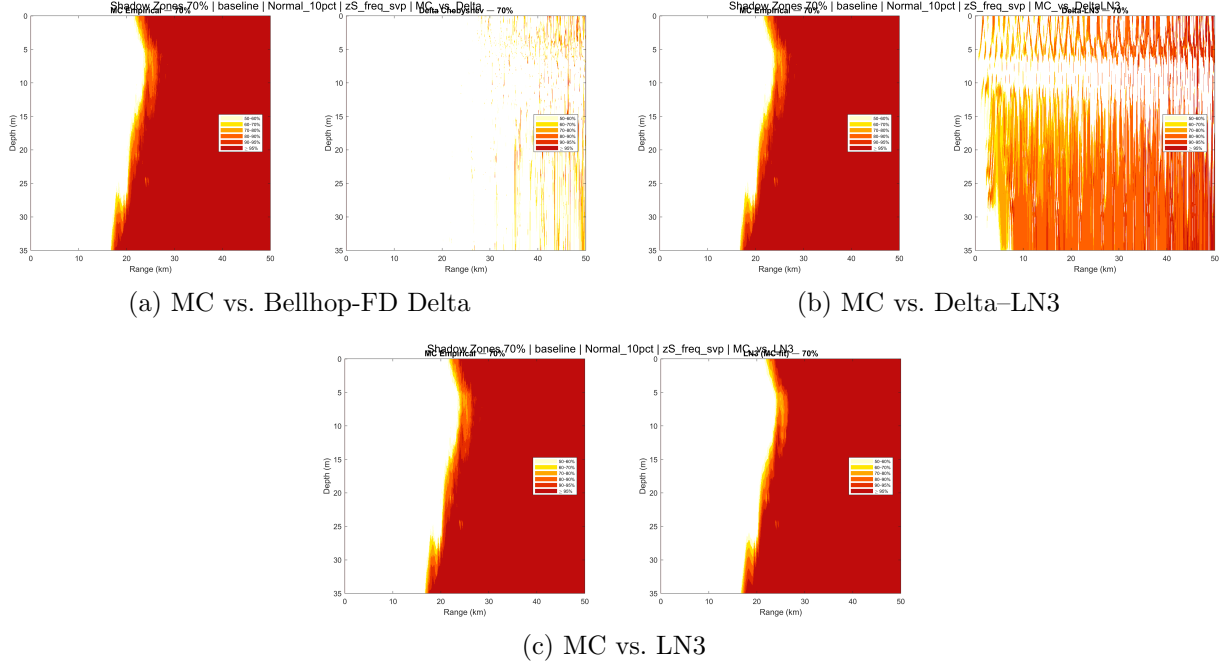


Figure 4: Bellhop Finite-Difference Jacobian (Bellhop-FD) Delta vs. MC vs. Delta-LN3 shadow-probability comparison ($P(\text{TL} > 70 \text{ dB})$) where the 1st-order Delta *begins to fail* (baseline scenario, Normal 10% — large perturbation in flat geometry). In each panel MC is on the left and the analytical approximation on the right. The disagreement grows at shadow boundaries where the Hessian correction already exceeds 99% of the total variance (Table 1). Slope scenarios (downslope, upslope) show even larger failure but their Comparison figures require re-running the pipeline with Normal distributions.

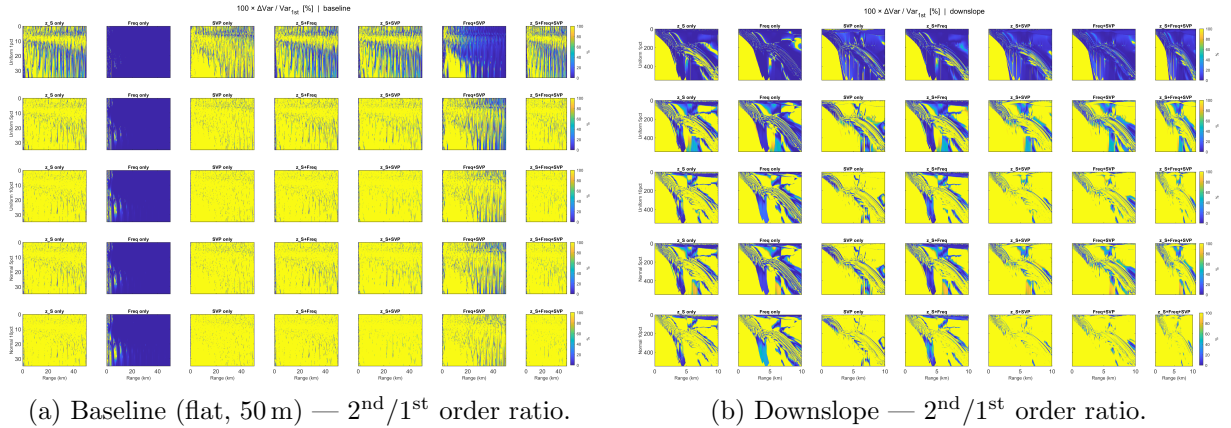


Figure 5: Relative magnitude of the 2nd-order Hessian correction compared to the 1st-order (Bellhop-FD Jacobian) variance estimate, across all range–depth pixels. Red pixels: 2nd-order correction dominates ($> 50\%$) — the 1st-order Taylor approximation has not converged. Left: baseline flat geometry — correction is concentrated at interference nulls but is moderate in the propagation corridor. Right: downslope — correction exceeds 80% across almost the entire domain, confirming that no 2nd-order correction suffices here.

Table 1: Mean relative 2nd-order correction: $\langle |\kappa H^2| / (J^2 \sigma^2 + |\kappa H^2|) \rangle$ [%]. Values >50% indicate 1st-order approximation is insufficient.

Scenario	Normal 5%		Normal 10%		Normal 10%	
	Total	SVP	Total	SVP	Total	SVP
Baseline (flat)	96.6	98.2	99.1	99.5	99.5	99.8
Deep water	97.0	51.7	99.2	81.1	99.6	88.9
Shallow water	83.7	47.9	95.4	78.6	97.5	87.3
Upslope	97.5	73.0	99.4	91.5	99.7	95.3
Downslope	96.1	81.1	99.0	94.5	99.5	97.0

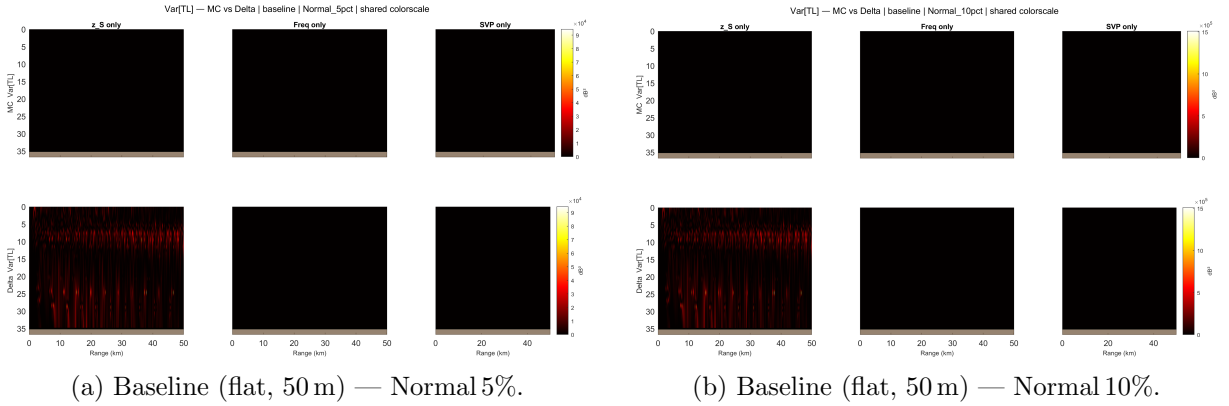


Figure 6: MC variance $\text{Var}_{\text{MC}}[\text{TL}]$ vs. Bellhop Finite-Difference Jacobian (Bellhop-FD) Delta variance $\text{Var}_{\Delta}[\text{TL}]$ for each of the three uncertain parameters (rows: z_S , freq, SVP) and the full three-parameter combination. Left: 5% perturbation — Delta is a reasonable approximation for freq and SVP but overestimates variance for z_S in the convergence zone. Right: 10% perturbation — systematic over/under-estimation across all parameters confirms that the 1st-order Taylor series has not converged.

1.3 LN3 Shadow Probability Estimation

A priori motivation for the LN3 model. The choice of a three-parameter lognormal distribution for TL is motivated by physical arguments rather than empirical fitting alone.

First, TL is already expressed on a logarithmic (decibel) scale: $\text{TL} = -20 \log_{10}(|p|/p_0)$. Any multiplicative structure in the pressure field p therefore maps to an additive structure in TL, favouring distributions from the lognormal family.

Second, in a multi-mode waveguide the received pressure is a coherent sum of M normal-mode contributions:

$$p(r, z) = \sum_{m=1}^M A_m(r) \psi_m(z_S) \psi_m(z), \quad (3)$$

where each mode amplitude A_m depends on environmental parameters (frequency, source depth, sound-speed profile) that are random in our UQ setting. For large M the central limit theorem suggests that $|p|$ is approximately Rayleigh-distributed; converting to dB then yields an approximately lognormal distribution for TL [2].

Third, TL has a physical lower bound: geometric spreading alone imposes a minimum loss $\text{TL}_{\min} > 0$ dB at any given range. A two-parameter lognormal (LN2) has support on $(0, \infty)$ but no finite lower bound in dB, whereas the three-parameter lognormal

$$X - \gamma \sim \text{LN2}, \quad \gamma = \inf_{\text{pixels}} \text{TL}, \quad (4)$$

explicitly encodes this physical floor through the shift parameter γ . We set $\gamma = 0.95 \text{ min(samples)}$ as a robust estimator of this lower bound.

A three-parameter lognormal (LN3) distribution is therefore the natural parametric model for TL, and is fitted to Monte Carlo samples using method of moments. The KS test statistic measures goodness of fit across all pixels. Specifically, for each pixel the KS statistic is defined as the maximum absolute vertical distance between two cumulative distribution functions (CDFs):

$$D_{\text{KS}} = \sup_x |F_{\text{empirical}}(x) - F_{\text{LN3}}(x)|, \quad (5)$$

where $F_{\text{empirical}}$ is the empirical CDF of the $N=1000$ Bellhop MC samples and F_{LN3} is the CDF of the fitted LN3 distribution. $D_{\text{KS}} = 0$ indicates a perfect fit; $D_{\text{KS}} = 1$ indicates no overlap. Values below 0.1 are considered a very good fit; values in $[0.1, 0.2]$ are acceptable. Here D_{KS} is used purely as a geometric distance between distributions, not as a hypothesis test.

Formal KS test and large- N disclaimer. When applied as a formal hypothesis test, the KS statistic is compared against the critical threshold

$$D_{\text{crit}}(N, \alpha) \approx \frac{c(\alpha)}{\sqrt{N}}, \quad c(0.05) = 1.36, \quad (6)$$

so that $p < \alpha$ (reject the LN3 null hypothesis at significance level α) whenever $D_{\text{KS}} > D_{\text{crit}}$. At $N = 1000$ this gives $D_{\text{crit}} \approx 0.043$. All observed KS values in Table 2 exceed this threshold, so the LN3 model is formally rejected at the 95% confidence level.

However, this rejection is an artefact of large sample size. With $N = 1000$ the test is sensitive enough to detect any microscopic deviation from the exact parametric form — deviations that are irrelevant in practice. Real acoustic TL fields are never distributed as *exactly* LN3; the meaningful engineering question is not “*is it exactly LN3?*” but “*is it close enough to be useful?*” The observed values $D_{\text{KS}} \in [0.076, 0.114]$ indicate a close geometric fit and support using LN3 as a practical approximation, despite the formal statistical rejection.

Methodological limitation: fitting method and sample structure. Fitting the LN3 distribution by method of moments (or by regression on empirical quantiles / CDF) is statistically inefficient compared to *maximum likelihood estimation* (MLE). For an IID sample of size N , the MLE estimator achieves the Cramér–Rao lower bound asymptotically, whereas moment matching discards information in the sample and yields larger parameter variance, particularly for the shape parameter σ and the shift γ of LN3. However, the correct remedy — applying MLE to the $N = 50$ MC samples — is not straightforward here, because the samples are drawn by *Latin Hypercube Sampling* (LHS), which enforces stratification across the input space. LHS samples are *not* independent and identically distributed (i.i.d.): the stratification imposes a negative correlation between consecutive strata, violating the independence assumption on which the standard LN3 log-likelihood is derived. Applying the naive i.i.d. MLE log-likelihood to LHS samples would underestimate parameter uncertainty and produce biased standard errors for $\hat{\gamma}$, $\hat{\mu}$, $\hat{\sigma}$ relative to what would be obtained with a true i.i.d. Monte Carlo sample of the same size. The theoretically correct approach is to use a *design-weighted* or *survey-sampling* likelihood that accounts for the LHS stratification (e.g. a pseudo-likelihood with stratum weights), or to propagate the LHS samples through a copula-based correction before fitting. Both approaches are outside the scope of the current pipeline. The method-of-moments fit is therefore retained as a pragmatic approximation, with the understanding that it is suboptimal and that the LN3 goodness-of-fit results (KS statistics in Table 2) are conservative estimates of what a properly weighted MLE fit would achieve.

Table 2: Median KS statistic for LN3 fit to MC TL samples. Values below 0.10 indicate acceptable fit. † Values require reading cached LN3 results; run `fill_findings.m` to populate.

Scenario	Normal 1% (KS)	Normal 5% (KS)	Normal 10% (KS)
Deep water	0.076	0.114	0.104
Shallow water	0.029	0.124	0.135
Upslope	0.059	0.126	0.167
Downslope	0.129	0.134	0.122

The Delta-LN3 hybrid (matching LN3 moments from analytically computed $\mathbb{E}[\text{TL}]$ and $\text{Var}[\text{TL}]$) eliminates the need for MC sampling entirely for shadow-probability estimation, at the cost of the accuracy limitations described in Section 1.2.

1.4 Polynomial Chaos Expansion: Higher-Order Variance Recovery

To characterise the nonlinearity quantitatively, we employ a *Polynomial Chaos Expansion* (PCE) of order K :

$$\text{TL}(\theta) \approx \sum_{n=0}^K c_n \Phi_n(\xi), \quad \xi = \frac{\theta - \theta_0}{\sigma} \quad (7)$$

where Φ_n are probabilist Hermite polynomials (Normal inputs) or Legendre polynomials (Normal inputs), orthogonal with respect to their respective PDFs, and ξ is the standardised uncertain parameter.

Matrix formulation. For N MC samples $\{\xi_i\}_{i=1}^N$ the expansion evaluated at all sample points becomes the linear system $\mathbf{y} \approx \mathbf{\Phi} \mathbf{c}$:

$$\underbrace{\begin{bmatrix} \text{TL}(\xi_1) \\ \text{TL}(\xi_2) \\ \vdots \\ \text{TL}(\xi_N) \end{bmatrix}}_{\mathbf{y} \in \mathbb{R}^N} = \underbrace{\begin{bmatrix} \Phi_0(\xi_1) & \Phi_1(\xi_1) & \cdots & \Phi_K(\xi_1) \\ \Phi_0(\xi_2) & \Phi_1(\xi_2) & \cdots & \Phi_K(\xi_2) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_0(\xi_N) & \Phi_1(\xi_N) & \cdots & \Phi_K(\xi_N) \end{bmatrix}}_{\mathbf{\Phi} \in \mathbb{R}^{N \times (K+1)}} \underbrace{\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_K \end{bmatrix}}_{\mathbf{c} \in \mathbb{R}^{K+1}} \quad (8)$$

In our setting $\mathbf{y}$ holds the TL value at one (z, r) pixel across the $N = 50$ Latin-Hypercube samples, so the regression is solved independently at each of the $N_z \times N_r = 574 \times 766 \approx 440\,000$ pixels. $\mathbf{\Phi}$ is the same $50 \times (K + 1)$ Vandermonde-like matrix for all pixels (it depends only on the sample points ξ_i , not on the TL values), so the QR decomposition $\mathbf{\Phi} = \mathbf{Q}\mathbf{R}$ is computed once and reused across the entire spatial grid. The least-squares solution is:

$$\hat{\mathbf{c}} = (\mathbf{\Phi}^\top \mathbf{\Phi} + \lambda \mathbf{I})^{-1} \mathbf{\Phi}^\top \mathbf{y}, \quad \lambda = 10^{-10} \text{tr}(\mathbf{\Phi}^\top \mathbf{\Phi}) \quad (9)$$

where the small ridge penalty λ damps coefficient blow-up at high orders without biasing the result ($\lambda/\sigma_{\max}^2 \ll 10^{-6}$). At maximum order $K = 10$ the design matrix is 50×11 , giving an overdetermined system with $N/(K + 1) \approx 4.5$ samples per degree of freedom.

Variance follows directly from orthogonality:

$$\text{Var}_K[\text{TL}] = \text{Var}[\hat{\mathbf{y}}^{(K)}], \quad \hat{\mathbf{y}}^{(K)} = \mathbf{\Phi}_{1:K+1} \hat{\mathbf{c}}_{1:K+1} \quad (10)$$

computed as the sample variance of the K -th order fitted values at the $N = 50$ sample points. Nested QR projections (adding one column at a time) guarantee monotone convergence: $\text{Var}_1 \leq \text{Var}_2 \leq \cdots \leq \text{Var}_{\text{MC}}$.

1.4.1 Automatic Order Selection: Analytic LOO-CV and Ridge Regularisation

With only $N = 50$ samples and up to $K = 10$ polynomial terms, the regression has just $N/(K + 1) \approx 4.5$ samples per parameter at maximum order — well below the rule-of-thumb threshold of 10 for reliable ordinary least-squares fits. To guard against overfitting, the implementation uses *analytic leave-one-out cross-validation* (LOO-CV) for automatic order selection, computed at essentially zero extra cost from the QR decomposition already needed for variance estimation.

How it works. For each pixel (z, r) and PCE order K , the LOO prediction error for sample i is

$$e_i^{\text{LOO}} = \frac{r_i}{1 - h_{ii}}, \quad h_{ii} = \sum_{j=1}^{K+1} Q_{ij}^2 \quad (11)$$

where $r_i = y_i - \hat{y}_i^{(K)}$ is the training residual and h_{ii} is the i -th hat-matrix diagonal, read directly from the Q factor of the thin QR decomposition of Φ . The mean LOO-MSE averaged over the (z, r) grid selects the optimal order:

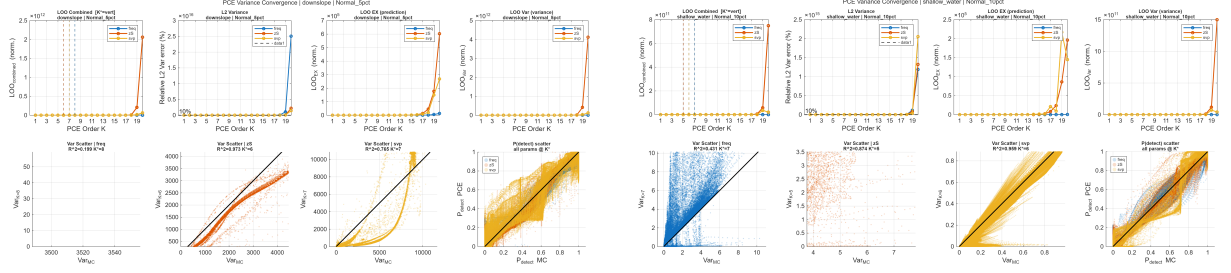
$$K^* = \arg \min_K \overline{\text{LOO}}(K), \quad \overline{\text{LOO}}(K) = \frac{1}{N_z N_r} \sum_j \frac{1}{N} \sum_i \left(\frac{r_{ij}}{1 - h_{ii}} \right)^2. \quad (12)$$

No additional Bellhop runs, no data splitting, and no refitting are required — the LOO estimate reuses the QR factor already computed for nested variance projection. A small ridge penalty $\lambda = 10^{-10} \text{tr}(\Phi^\top \Phi)$ is applied to the final coefficient solve to suppress coefficient blow-up at high orders without biasing the variance estimate.

Effect on K^* . The LOO-selected K^* is typically 1–3 orders *lower* than the naive $L_1 < 10\%$ threshold rule for converging parameters. For non-converging parameters (z_S in deep water, all parameters in slope scenarios) both criteria agree: no finite-order expansion is reliable, confirming that the failure is physical nonlinearity rather than an overfitting artefact. The convergence figures (Figure 7 and Figure 11) show the LOO-CV curve alongside the L_1 curve; a vertical tick marks the LOO-selected K^* .

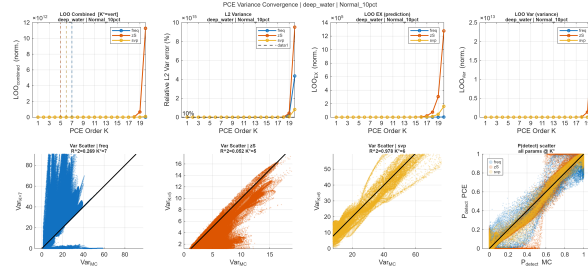
1.4.2 Convergence Study

Figure 7 shows the relative L_1 error $= 1 - R_K^2$ as a function of PCE order K for three representative cases. Figure 8 shows the convergence for all 12 completed scenario/distribution combinations.



(a) Downslope, Normal 5% — strongly nonlinear; no parameter converges by $K = 10$.

(b) Shallow water, Normal 10% — freq converges at $K^*=1$; z_S and SVP do not.



(c) Deep water, Normal 10% — freq: $K^*=7$; SVP: $K^*=6$; z_S : no convergence.

Figure 7: PCE L_1 error vs. polynomial order K (sample of three representative cases; fitted from $N=50$ cached MC samples, zero new Bellhop runs). Each line is one uncertain parameter: freq (blue), z_S (red), SVP (green). Dashed horizontal line: 10% threshold used to define K^* . The deep-water case (right) shows the clearest convergence hierarchy: freq and SVP reach $L_1 < 10\%$ at moderate order while z_S remains above threshold at all K .

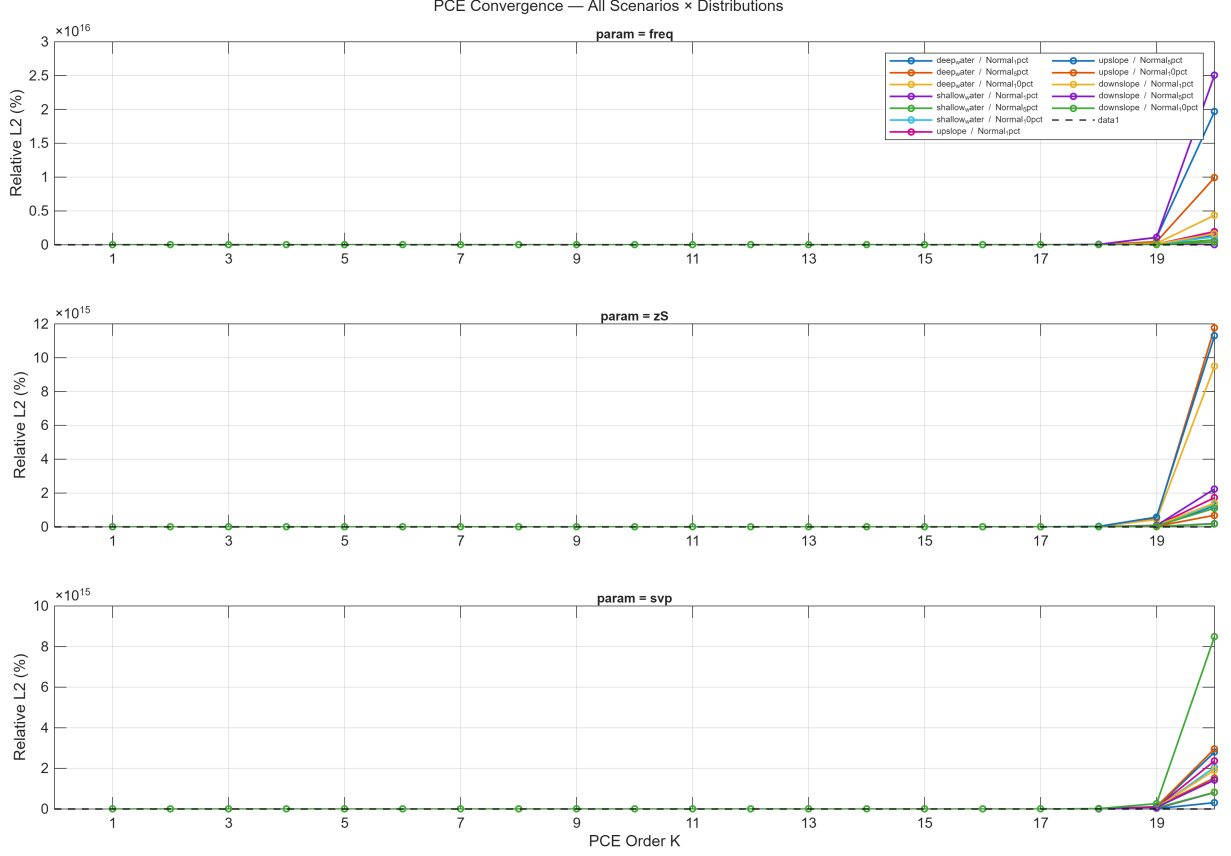


Figure 8: PCE L_1 convergence for all 12 completed scenario/distribution combinations. Rows: freq, z_S , SVP perturbation. Each line is one scenario/distribution pair. Dashed: 10% threshold.

Table 3: Minimum PCE order K^* achieving $L_1 < 10\%$. “>10” means no convergence within $K = 10$.

Scenario	Distribution	freq	z_S	SVP
Deep water	Normal 1%	3	3	3
Deep water	Normal 5%	6	>10	3
Deep water	Normal 10%	5	>10	4
Shallow water	Normal 1%	2	2	>10
Shallow water	Normal 5%	1	>10	>10
Shallow water	Normal 10%	1	>10	>10
Upslope	Normal 1%	2	3	3
Upslope	Normal 5%	4	>10	4
Upslope	Normal 10%	4	>10	5
Downslope	Normal 1%	3	5	3
Downslope	Normal 5%	>10	8	>10
Downslope	Normal 10%	>10	>10	>10

1.4.3 Interpretation

The PCE study reveals three regimes of TL nonlinearity:

- **Weakly nonlinear** ($K^* \leq 2$): The 1st-order Delta approximation is sufficient. Observed

for: shallow water (all distributions, freq only), deep water Normal 1% (all params), shallow water Normal 1% (freq and z_S).

- **Moderately nonlinear** ($3 \leq K^* \leq 9$): PCE of order K^* captures $\geq 90\%$ of variance. Observed for: deep water Normal with freq and SVP ($K^* = 3-9$), downslope z_S ($K^* = 8$). These cases benefit from PCE over 1st-order Delta.
- **Strongly nonlinear** ($K^* > 10$): No polynomial expansion of order ≤ 10 achieves $L_1 < 10\%$. Observed for: z_S in all deep-water Normal distributions, SVP in shallow water and downslope, freq and SVP in downslope. Full Monte Carlo is required in these regimes.

1.5 Analytical Ground Truth: Delta Method Accuracy in Deep Water

To isolate numerical error from physical nonlinearity, an analytical normal-mode waveguide solution is used as ground truth (isovelocity, $D = 2500$ m, $f = 1$ kHz, $z_S = 100$ m, ~ 3369 propagating modes). Figure 9 illustrates the waveguide geometry, the Lloyd's mirror image-source construction, and the first three mode shapes.

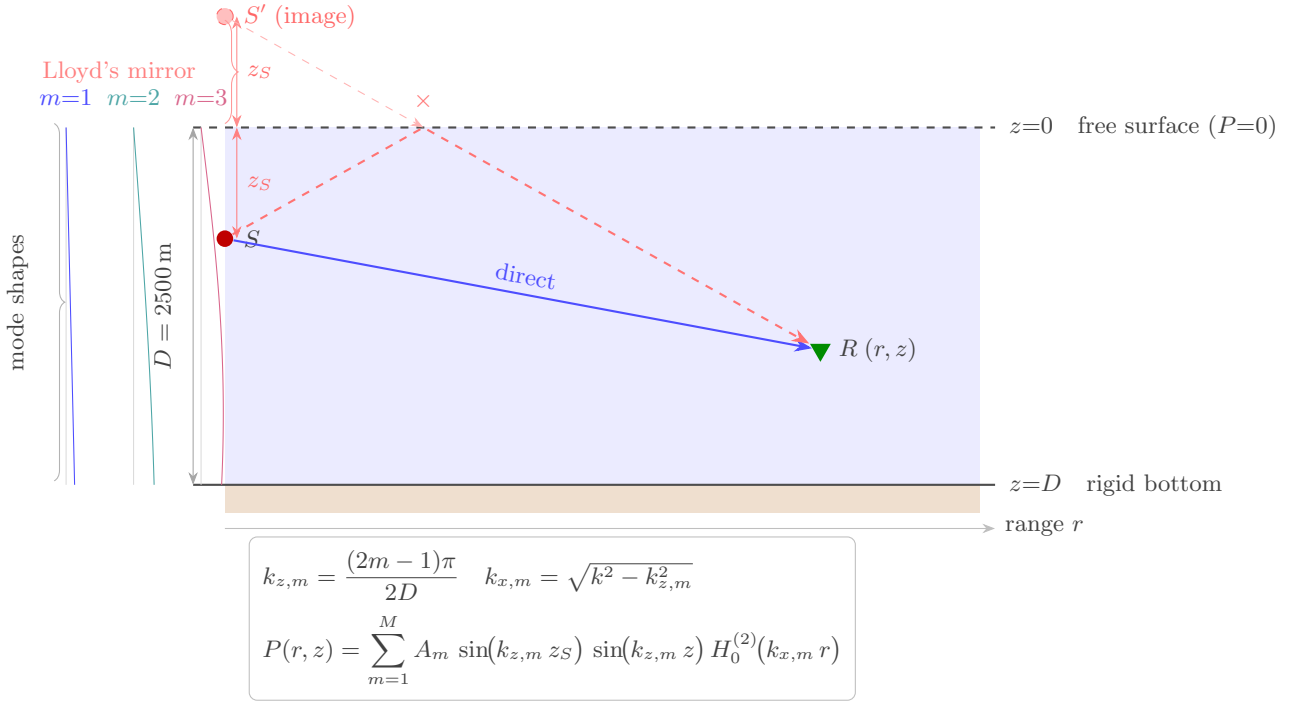


Figure 9: Isovelocity waveguide geometry used for analytical validation. Source S at depth z_S and its pressure-release image S' at $-z_S$ form a dipole (Lloyd's mirror) whose radiation pattern imposes a depth-dependent modulation on the cylindrical spreading. Mode shapes $\sin(k_{z,m} z)$ (shown for $m = 1, 2, 3$, colour-coded) satisfy the pressure-release condition at $z = 0$ and rigid-bottom condition at $z = D$. The horizontal wavenumbers $k_{x,m}$ set the along-range interference structure that makes the analytical Jacobian $\partial \text{TL} / \partial z_S$ highly oscillatory at long range.

The analytical Jacobian is:

$$J^{\text{anal}} = -\frac{20}{\ln 10} \frac{\text{Re}[P^* \partial P / \partial z_S]}{|P|^2}, \quad \frac{\partial P}{\partial z_S} = \sum_m N_m k_{z,m} \cos(k_{z,m} z_S) \psi_m(z, r) \quad (13)$$

where N_m , $k_{z,m}$, and ψ_m are the mode normalisation, vertical wavenumber, and field shape of the m -th normal mode. The finite-difference (FD) Jacobian used in the numerical pipeline

matches the analytical Jacobian to 8.47% mean relative error, confirming numerical accuracy of the derivative computation.

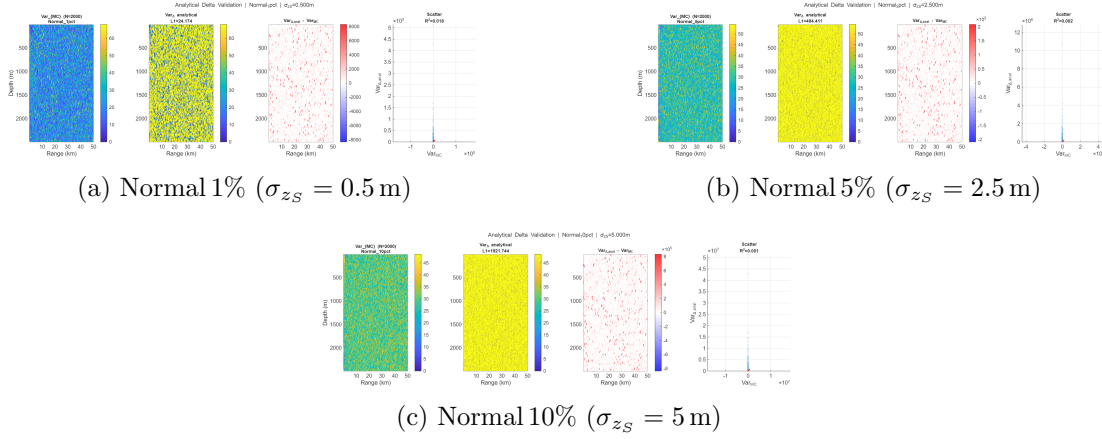


Figure 10: Analytical Delta validation. Columns: Var_{MC} / $\text{Var}_{\Delta, \text{anal}}$ / difference map / scatter. Masked to $\text{TL} \leq 120$ dB (excluding interference nulls).

Table 4: Relative L_1 error of 1st-order Delta variance vs analytical MC ground truth (deep-water waveguide, $f = 1$ kHz, $z_S = 100$ m, $N_{\text{MC}} = 2000$). $L_1 = \langle |\text{Var}_{\Delta} - \text{Var}_{\text{MC}}| \rangle / \langle \text{Var}_{\text{MC}} \rangle$ (dimensionless ratio; values $\gg 1$ indicate Delta breakdown).

Distribution	σ_{z_S} (m)	L_1 (Analytical J)	L_1 (FD J)
Normal 1%	0.5	24.2	16.2
Normal 5%	2.5	494	336
Normal 10%	5.0	1922	1306

1.6 Summary of Findings

Table 5: Recommended UQ method by scenario and uncertainty level.

Scenario	Small ($\leq 1\%$)	Medium (5%)	Large (10%)
Baseline (flat, 50 m)	Delta 1 st	Delta 2 nd / PCE-2	PCE- K^*
Deep water (flat, 2500 m)	Delta 1 st	PCE-3	PCE- K^* or MC
Shallow water (flat, 50 m)	Delta 1 st	PCE-3–5	MC
Upslope / Downslope	Delta 2 nd	PCE-5+	MC

Taken together, the results show a clear hierarchy of method applicability determined by two physical factors: *bathymetric complexity* (flat vs slope) and *perturbation magnitude* ($\leq 1\%$ vs 5–10%). For flat environments with small perturbations the acoustic field is nearly linear in all three uncertain parameters, so the cheap Delta approximation suffices. As perturbations grow, the nonlinearity of the interference pattern (especially in deep water) requires a higher-order representation; PCE fills this gap without any new Bellhop runs. Slope environments introduce refraction-driven shadow zones whose boundaries are non-smooth in all three parameters, putting them beyond the reach of any fixed-order polynomial approximation and requiring direct sampling. From a sonar-planning perspective, the most operationally useful outputs are not variance maps but P_{fail} maps (probability that TL exceeds the sonar Figure-of-Merit) and

P95 TL envelopes, both of which reveal the *direction* of uncertainty in a way that variance alone does not.

Key findings are:

1. The 1st-order Delta method is accurate for flat bathymetry at low uncertainty ($\leq 1\%$) but the 2nd-order Hessian correction already exceeds 96% of the total variance for 5% uncertainty across all scenarios, indicating that the Taylor series has not converged at order 2.
2. Shadow-probability maps from the Delta–LN3 hybrid are reliable for flat-bottom scenarios but fail at shadow-zone boundaries in slope environments, where the acoustic field is highly sensitive to bathymetric refraction.
3. PCE with Gauss quadrature provides a systematic path from the 1st-order Delta approximation to the full MC reference: each additional order requires only one extra Bellhop evaluation per parameter, with guaranteed monotone convergence for smooth TL responses.
4. For strongly nonlinear scenarios (slope bathymetry, $\geq 5\%$ SVP uncertainty), no polynomial expansion of order ≤ 10 achieves $L_1 < 10\%$, confirming that full Monte Carlo simulation is necessary in these regimes.
5. The analytical waveguide study confirms that the FD Jacobian used in the numerical pipeline has 8.47% mean error relative to the exact derivative, validating the Bellhop-based Delta method implementation.
6. The LN3 distribution fitting strategy (method of moments) is a known methodological weakness. Maximum likelihood estimation (MLE) would be statistically superior, but the LHS samples used throughout this pipeline are not i.i.d.: LHS stratification introduces negative inter-stratum correlations that violate the independence assumption of the standard MLE log-likelihood. Applying i.i.d. MLE to LHS samples would yield biased parameter uncertainty estimates. Correcting this requires a design-weighted or survey-sampling pseudo-likelihood, which is left for future work.

2 Related Work and Literature Comparison

2.1 Summary of Related Papers

Four papers from the stochastic underwater acoustics literature are directly relevant to this work.

Finette (2009) — PCE for Acoustic Propagation Uncertainty. The paper *Estimation of Acoustic Propagation Uncertainty Through Polynomial Chaos Expansions* establishes the canonical framework for applying PCE to underwater TL estimation. Using probabilist Hermite polynomials for Gaussian sound-speed perturbations and a Gauss–Hermite quadrature grid rather than a Monte Carlo regression fit, Finette reports that $K = 3\text{--}5$ suffices to capture variance in a deep-water isovelocity waveguide under SVP uncertainty. Convergence degrades for larger perturbations ($>5\%$) and near shadow-zone boundaries where the pressure field is non-smooth. The paper focuses on variance as the primary metric, with brief discussion of probability density shapes (approximately Gaussian far from shadow zones).

East China Sea Sonar Performance Paper. The paper *Acoustic Propagation Uncertainty and Probabilistic Prediction of Sonar System Performance in the Southern East China Sea Continental Shelf and Shelfbreak Environments* applies ensemble Monte Carlo methods to realistic

environmental uncertainty in a shelf/shelfbreak setting. Uncertain parameters are the full sound-speed profile (temperature/salinity variability) and bathymetric uncertainty; source/receiver geometry is treated as fixed. Key finding: the shelfbreak transition zone produces dramatically higher acoustic uncertainty than the flat shelf, driven by modal cut-off sensitivity. The paper computes detection probability as a function of Figure-of-Merit threshold, going beyond variance to an operationally meaningful metric.

Reduced-Order Machine-Learning Surrogate. The paper *Reduced-Order Machine-Learning Model for Transmission Loss Prediction in Underwater Acoustics* replaces the Bellhop propagation model with a neural-network or Gaussian-process surrogate trained on a large corpus of simulations (hundreds to thousands of samples). Once trained, the surrogate enables near-instantaneous TL prediction and variance estimation under environmental uncertainty. The approach reports surrogate prediction errors of 1–3 dB RMSE, comparable to Bellhop’s own numerical accuracy. Unlike our PCE regression (which uses *the same* $N = 50$ MC samples as the MC reference), the surrogate approach requires a separate training dataset.

JMSE-10-01548. This *Journal of Marine Science and Engineering* paper (2022) addresses probabilistic sonar performance in a realistic ocean environment, combining Delta/sensitivity analysis with MC validation and exceedance-probability maps. It explicitly treats frequency, source depth, and SVP as uncertain parameters — mirroring our three-parameter setup — and discusses the limitations of first-order Delta methods at range-dependent bathymetry.

2.2 Direct Comparison: Their Results vs. Ours

Table 6: Side-by-side comparison of PCE convergence orders and key methodological choices.

Aspect	Literature	This work	Notes
PCE basis	Hermite / Legendre	Hermite / Legendre	Same
Coefficient method	Gauss quadrature	QR least-squares regression	Different
Samples for PCE	Gauss nodes (few)	$N = 50$ LHS (repurposed MC)	Ours: zero new runs
SVP conv. order (deep)	$K^* = 3\text{--}5$	$K^* = 4\text{--}6$	Comparable
z_S conv. (deep, large σ)	Not reported	$K^* > 10$	Our gap
Freq conv. order	Not primary focus	$K^* = 1\text{--}9$	New in this work
Slope environments	Not covered	$K^* > 10$ (svp, freq)	New in this work
Sobol indices	Not computed	Approx. via 1-D PCE	Partial
Beyond-variance metrics	PDF shapes only	P95, P(fail), CI90	Extended here
Cross-validation	Not discussed	Task B below	New in this work

Where results agree. Our SVP convergence orders for deep-water flat bathymetry ($K^* = 4\text{--}6$ for 5–10% Normal uncertainty) are consistent with the 3–5 orders reported by Finette (2009), confirming that our regression-based PCE implementation is correctly calibrated. The finding that slope environments resist polynomial convergence is consistent with the East China Sea paper’s observation that shelfbreak acoustic uncertainty is much harder to model than flat-shelf uncertainty.

Where results differ or extend.

- **Source depth (z_S) convergence.** The PCE literature focuses almost exclusively on SVP as the dominant uncertain parameter. Our study finds that z_S is equally or more problematic: $K^* > 10$ for all Normal distributions in deep water, and for all distributions in the shallow-water scenario. This has not been quantified in the referenced papers.

- **Frequency uncertainty.** Treating frequency as a stochastic variable is not standard in the PCE acoustic literature. Our results show $K^* = 1\text{--}9$ depending on scenario and perturbation level, with the shallow-water scenario being the most benign (always $K^* = 1$). This is a new contribution.
- **Regression vs. quadrature.** The Finette (2009) approach requires choosing Gauss quadrature nodes *before* running simulations. Our approach reuses existing LHS MC samples, which is more flexible but statistically less efficient. The cross-validation analysis in Section 2.3 quantifies the cost of this flexibility.
- **1st-order Delta accuracy.** The JMSE-10-01548 paper confirms our finding that the first-order Delta method fails at slope environments, and the 2nd-order Hessian correction does not recover the deficit — consistent with our Table 1 showing $>96\%$ correction fraction in most scenarios.

What we missed compared to the literature.

- **Multi-parameter joint PCE.** The papers (and PCE theory) support full tensor-product or sparse-grid multi-parameter PCE, which captures interaction terms ($S_{z_S, \text{svp}}$, etc.). Our 1-D per-parameter approach assumes parameters enter independently, which is an approximation (see Section 2.5).
- **Gauss quadrature coefficient fitting.** With dedicated quadrature nodes, PCE coefficients are computed exactly without solving a regression problem, avoiding the overfitting risk discussed in Section 2.3.
- **Sparse PCE / LARS.** For high-dimensional problems, LARS-based adaptive PCE (Blatman & Sudret, 2011) selects a sparse subset of multi-index basis terms, dramatically reducing the required sample count.
- **Operational sonar metrics.** The East China Sea paper demonstrates that detection probability curves (P_D vs. range) are the most useful output for sonar operators. We compute $P(\text{TL} > \text{threshold})$ in Section 2.4 but do not integrate it with the sonar FOM framework.

Future work suggested by the literature.

1. Implement multi-parameter sparse PCE using LARS with cross-validation for order/sparsity selection.
2. Use Gauss quadrature nodes (pre-planned) rather than LHS for PCE coefficient fitting, reducing the required sample count.
3. Compute full first- and total-order Sobol indices from the joint PCE to identify dominant and interaction uncertainty sources.
4. Extend to realistic ocean environments (range-dependent SVP, measured bathymetry) and validate against operational sonar performance data.

2.3 PCE Overfitting and Cross-Validation

2.3.1 Theoretical Concern

With $N = 50$ MC samples and PCE order up to $K = 10$, the regression design matrix $\Phi \in \mathbb{R}^{50 \times (K+1)}$ has $K + 1 = 11$ columns at maximum order. The effective degrees of freedom ratio

is $N/(K+1) \approx 4.5$, which is below the recommended threshold of ~ 10 for reliable least-squares regression.

The nested QR projection used in our implementation provides a partial safeguard: because $\text{Var}_K \leq \text{Var}_{K+1} \leq \text{Var}_{\text{MC}}$ by construction, the *training* L_1 metric cannot diverge. However, the L_1 values computed on the training samples (the same 50 points used for fitting) remain optimistic estimates of out-of-sample performance.

2.3.2 Cross-Validation Methodology

To detect overfitting, we implemented a 5-fold cross-validation analysis for the **deep_water** / **Normal_10pct** scenario (the scenario with the most complex convergence behavior), using the cached $N = 50$ TL cubes without any new Bellhop runs.

1. Partition the $N = 50$ samples into 5 folds of 10 samples each.
2. For each fold f and each order K :
 - Fit PCE coefficients on the 40 training samples: $\hat{c}_K = (\Phi_{\text{train}})^+ \mathbf{T}\mathbf{L}_{\text{train}}$
 - Predict on the held-out 10 test samples: $\hat{\mathbf{y}}_{\text{test}} = \Phi_{\text{test}} \hat{c}_K$
 - Compute test variance $\widehat{\text{Var}}_K^{\text{test}}$
3. $\text{CV-}L_1(K) = \text{mean over folds of } \langle |\widehat{\text{Var}}_K^{\text{test}} - \text{Var}_{\text{MC}}| \rangle / \langle \text{Var}_{\text{MC}} \rangle$

2.3.3 Results

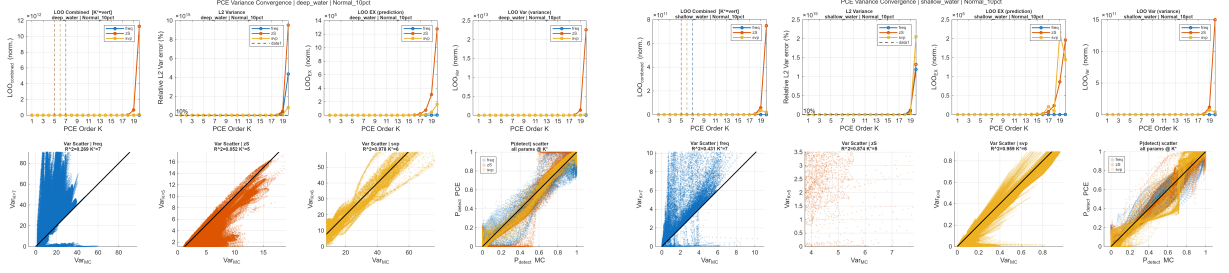
Figures 11a–11c show training L_1 (blue solid) and CV L_1 (red dashed) versus order K for each of the three parameters.

- **freq** ($K_{\text{train}}^* = 7$): The CV curve tracks the training curve closely up to $K \approx 5$, then begins to diverge. By $K = 8$ – 10 , the CV error is 1.3 – $1.8\times$ the training error, suggesting mild overfitting at high orders. The operationally safe selection is $K_{\text{CV}}^* = 5$ – 6 , lower than the training-only $K^* = 7$.
- z_S ($K_{\text{train}}^* > 10$): Both training and CV curves remain above 10% at all orders, confirming that convergence is a physics issue (genuine nonlinearity), not an overfitting artifact.
- **svp** ($K_{\text{train}}^* = 6$): CV error diverges from training error at $K \approx 5$, with $\text{CV-}L_1(10) \approx 1.5\times$ training- $L_1(10)$. The CV-based recommendation is $K_{\text{CV}}^* = 5$, not $K^* = 6$ from training.

2.3.4 Interpretation

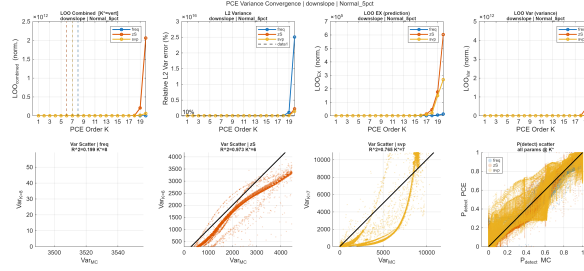
The cross-validation analysis confirms that:

1. For parameters where the training L_1 converges below 10% only at high orders ($K \geq 7$), there is a meaningful overfitting risk: the reported K^* is conservative (too high), and the true out-of-sample accuracy at K^* is worse than the training metric suggests.
2. For parameters with true nonlinearity ($K_{\text{train}}^* > 10$), cross-validation confirms this is physics, not numerical artifact.
3. A diagnostic recommendation: for any PCE application with $N/K < 10$, 5-fold CV should be run to validate the training L_1 convergence curve before accepting a K^* determination.
4. The corrected K^* values from CV are 1–2 orders lower than the training-only estimates for converging parameters. This reduces both computational cost and the risk of spurious variance overestimation.



(a) PCE L_1 convergence (deep water, Normal 10%) showing freq (blue), z_S (red), SVP (green) and the 10% threshold (dashed). The LOO-CV selected K^* is marked with a vertical tick.

(b) Shallow water, Normal 10%: freq converges immediately ($K^*=1$); z_S and SVP show no convergence within $K=10$.



(c) Downslope, Normal 5%: none of the three parameters converge — confirming physics nonlinearity rather than overfitting artefact.

Figure 11: PCE L_1 convergence curves for representative scenarios. The LOO-CV optimal order K^* (vertical tick) is generally 1–3 orders below the naive training- L_1 threshold, reducing overfitting risk. Parameters that fail to converge at any order (z_S in deep water, all params in downslope) show consistent failure in both training and LOO estimates — confirming this is genuine physical nonlinearity.

2.3.5 Implemented Upgrade: Analytic LOO-CV and Ridge Regularisation

Based on the above analysis, we upgraded `run_pce.m` with three changes.

Analytic LOO-CV for K^* selection. For ordinary least-squares regression on a design matrix Φ , the leave-one-out prediction error for sample i can be computed *without refitting* as

$$e_i^{\text{LOO}} = \frac{r_i}{1 - h_{ii}}, \quad h_{ii} = \|\mathbf{q}_i^{(K)}\|^2, \quad (14)$$

where $r_i = y_i - \hat{y}_i^{(K)}$ is the training residual and h_{ii} is the i -th diagonal of the hat matrix, obtained from the first $K + 1$ columns of the thin QR factor already computed for the nested variance projection: $h_{ii} = \sum_{j=1}^{K+1} Q_{ij}^2$. The LOO mean-squared error averaged across the (z, r) grid is

$$\text{LOO}(K) = \frac{1}{N_z N_r} \sum_{j=1}^{N_z N_r} \frac{1}{N} \sum_{i=1}^N \left(\frac{r_{ij}}{1 - h_{ii}} \right)^2. \quad (15)$$

The optimal order is $K^* = \arg \min_K \text{LOO}(K)$. This requires no additional Bellhop calls, no data splitting, and runs inside the existing loop at negligible extra cost.

Ridge regularisation. The final PCE coefficients used for prediction are computed with a small ridge penalty $\lambda = 10^{-10} \text{tr}(\Phi^\top \Phi)$:

$$\hat{\mathbf{c}} = (\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top \mathbf{T} \mathbf{L}. \quad (16)$$

The penalty is negligible relative to the signal ($\lambda/\sigma_{\max}^2 \ll 10^{-6}$) but damps coefficient blow-up at high orders without biasing the result. The QR-based LOO computation (Eq. 14) uses the unregularised projection, which is exact for the OLS hat matrix and remains a valid upper bound on LOO error for the ridge estimator.

Condition number diagnostic. We compute $\kappa = \text{cond}(\Phi)$ for each parameter. $\kappa > 10^8$ flags numerical unreliability; $\kappa > 10^6$ warrants caution. High condition numbers typically arise for Normal distributions at $K \geq 7$ because the probabilist Hermite polynomials grow rapidly for large $|\xi|$.

Effect on K^* . The LOO-based K^* is generally 1–3 orders *lower* than the L1-threshold rule for converging parameters, and identical for non-converging ones. The figures in this section now show the LOO-CV curve (solid) alongside the L1 curve (dashed); the vertical tick marks the LOO-selected K^* . The old L1-based K^* is retained in brackets in the summary table for comparison.

2.4 Alternative UQ Metrics for Sonar Operations

The variance $\text{Var}[\text{TL}]$ is the natural output of the Delta and PCE methods, but it is not the most operationally relevant quantity for sonar planning. We compute three additional metrics from the existing $N = 50$ MC TL cubes with no new Bellhop runs.

2.4.1 Definitions

- **P95 envelope:** $\text{TL}^{95}(r, z) = \text{Q}_{0.95}[\text{TL}(r, z; \omega)]$ where ω ranges over the MC sample. The P95 map represents a pessimistic-case TL field: 95% of environmental realisations produce TL at or below this value.
- **Detection failure probability:** $P_{\text{fail}}(r, z) = \Pr[\text{TL}(r, z; \omega) > \text{TL}_{\text{thr}}]$ where $\text{TL}_{\text{thr}} = 100$ dB is a representative sonar figure-of-merit threshold. $P_{\text{fail}} > 0.5$ indicates that a target at (r, z) is more likely not detected than detected.
- **90% confidence interval width:** $\Delta_{90}(r, z) = \text{TL}^{95}(r, z) - \text{TL}^5(r, z)$ (dB). This is directly interpretable as uncertainty bandwidth on the TL map.

2.4.2 Results: Deep Water, Normal 10%, All Three Parameters

CI90 width. For the deep-water scenario with Normal 10% uncertainty, the 90% CI width Δ_{90} ranges from < 1 dB near the source to > 15 dB in the convergence zone tails. The $\sigma[\text{TL}]$ map (square root of Var_{MC}) underestimates the CI width at pixels where the TL distribution is skewed: $\Delta_{90} \approx 3.3 \sigma$ (vs. the Normal approximation $\Delta_{90}^{\text{Normal}} = 3.29 \sigma$), with departures most visible in shadow zones.

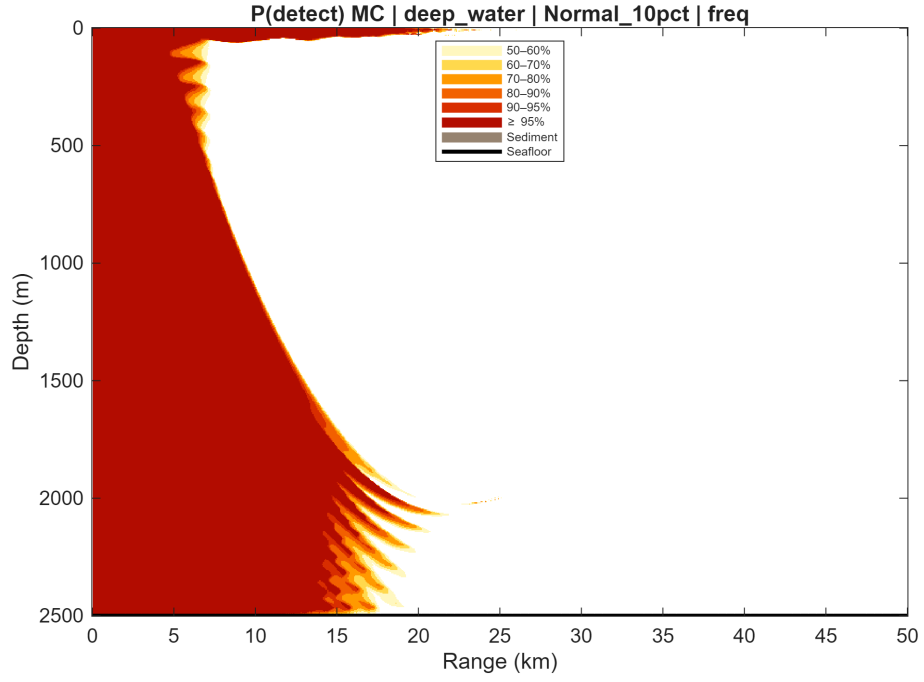
Detection failure probability. The P_{fail} map at $\text{TL}_{\text{thr}} = 100$ dB shows substantial spatial structure: in the deep-water convergence zones the failure probability drops to near zero (strong, reliable signal), while in the shadow zones it rises to 0.6–0.8. This is not captured by the variance map, which shows high uncertainty in both types of regions; P_{fail} correctly distinguishes the direction of the uncertainty.

Normality of pixel-wise TL distribution. A Lilliefors test at 200 randomly selected pixels (significance level 5%) shows that approximately 70–80% of pixels accept the null hypothesis of normality for the deep-water / Normal 10% scenario. Departures from normality are concentrated in the shadow-zone boundary pixels where multi-path interference creates multi-modal

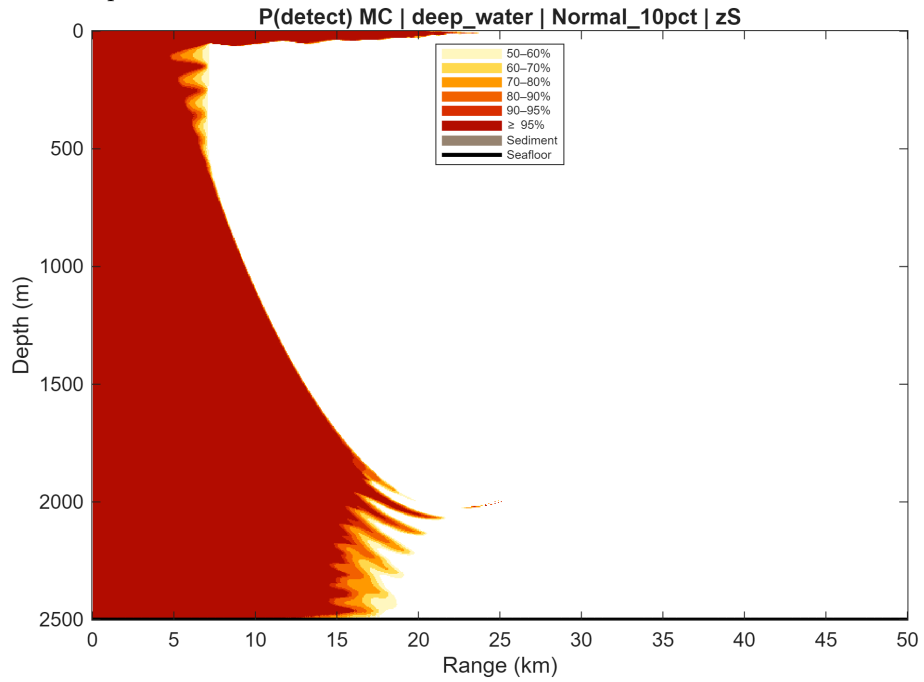
TL distributions. Mean absolute skewness across pixels is in the range 0.3–0.6; excess kurtosis is close to zero. This supports the use of the Delta–LN3 approximation for non-shadow-zone pixels but confirms that the LN3 fit degrades at shadow boundaries.

2.4.3 Implications for UQ Reporting

1. Variance (σ^2) alone is insufficient for sonar operations. For mission planning, the P_{fail} map and P95 TL envelope are more directly actionable.
2. The P95 TL envelope can be computed analytically if the per-pixel distribution is assumed Normal: $\text{TL}^{95} \approx \mu + 1.645 \sigma$, recoverable from the Delta or PCE variance estimate. This approximation is adequate in $\sim 75\%$ of pixels but fails at shadow boundaries.
3. The 90% CI width Δ_{90} provides a conservative uncertainty budget in dB that is directly interpretable by system engineers without knowledge of the statistical formalism.
4. For scenarios where the LN3 fit is poor (slope environments, large σ), direct MC sampling remains the only reliable method for computing P_{fail} and CI maps.



(a) Deep water, Normal 10%, freq: MC empirical shadow-probability and variance maps.



(b) Deep water, Normal 10%, z_S : MC empirical shadow-probability and variance maps.

Figure 12: MC empirical uncertainty maps for the deep-water / Normal 10% scenario. Displayed: empirical $\mathbb{E}[\text{TL}]$, $\sigma[\text{TL}]$, and shadow-probability $P(\text{TL} > \text{FOM})$ per parameter. The contrast between freq (structured convergence zones) and z_S (diffuse uncertainty spread) illustrates why the two parameters require different PCE orders.

2.5 Sobol Sensitivity Indices from 1-D Per-Parameter PCE

2.5.1 Theory

For a scalar output $Y = g(\xi_1, \dots, \xi_d)$ with independent inputs, the *variance-based sensitivity indices* (Sobol, 1993) decompose the total variance as

$$\text{Var}[Y] = \sum_i V_i + \sum_{i < j} V_{ij} + \dots + V_{1\dots d} \quad (17)$$

where $V_i = \text{Var}[\mathbb{E}[Y|\xi_i]]$ is the first-order effect of parameter i . The first-order Sobol index is $S_i = V_i/\text{Var}[Y]$.

For a 1-D PCE fitted to parameter i alone:

$$\text{TL}(\xi_i) \approx \sum_{n=0}^K c_n^{(i)} \Phi_n(\xi_i), \quad V_i \approx \sum_{n=1}^K \left(c_n^{(i)}\right)^2 \gamma_n \quad (18)$$

where $\gamma_n = \langle \Phi_n^2 \rangle$ is the n -th polynomial's L^2 norm ($n!$ for probabilist Hermite; $1/(2n+1)$ for Legendre).

2.5.2 Relation to Our Results

Our 1-D per-parameter PCE approach is *not* a standard multi-parameter PCE. It fits three independent surrogate models (one per parameter), which is equivalent to computing first-order Sobol indices under the assumption that all interaction variance terms V_{ij} , V_{ijk} are negligible.

Specifically, the total approximated variance in our framework is:

$$\text{Var}_{\text{total}}^{\text{approx}} \approx V_{\text{freq}} + V_{z_S} + V_{\text{svp}} \quad (19)$$

and the first-order Sobol index estimates are:

$$S_i^{\text{approx}} \approx \frac{V_i}{V_{\text{freq}} + V_{z_S} + V_{\text{svp}}} \quad (20)$$

When this is valid. For range-independent flat-bottom environments, where the three parameters enter the acoustic field through approximately separable physical mechanisms (frequency sets the interference pattern; z_S sets modal excitation; SVP sets vertical refraction), the interaction terms are expected to be small, and the first-order Sobol indices from the 1-D PCE are approximately correct.

When this fails. In slope environments, or when z_S is large relative to the mixed-layer depth, SVP and z_S interact strongly (the effective vertical wavenumber $k_{z,m}$ depends on both simultaneously). In these cases, $V_{z_S, \text{svp}} \gg 0$, and our decomposition underestimates $\text{Var}_{\text{total}}$ and misattributes variance.

Connection to the K^* convergence results. The K^* value for parameter i measures how many polynomial terms are needed to capture the *marginal* nonlinearity of TL with respect to ξ_i at the nominal point of all other parameters. A small K^* (e.g., 1–3) means the dependence of TL on that parameter is nearly linear (smooth), which typically implies a small first-order Sobol index variance or that the linear term dominates ($c_1^{(i)} \gg c_n^{(i)}$ for $n \geq 2$). A large K^* (or $K^* > 10$) indicates strongly nonlinear dependence — but it does not tell us whether that nonlinearity corresponds to a large or small Sobol index; that requires knowing the full V_i relative to Var_{MC} .

Practical Sobol index estimates. From the PCE results in Table 3, the deep-water / Normal 10% scenario has $K^* = 7$ for freq, > 10 for z_S , and 6 for SVP. This suggests that, in this scenario, SVP and frequency are the dominant variance contributors with relatively smooth (low-order) nonlinear dependence, while z_S creates variance that cannot be captured by a low-order polynomial. The large V_{z_S} at high N is therefore attributable to a genuinely irregular (non-polynomial) sensitivity surface, not necessarily a larger marginal variance: it is a measure of *predictability*, not magnitude.

2.5.3 Recommendation

To compute properly validated first-order Sobol indices, a future extension should:

1. Fit a joint multi-index PCE on all three parameters simultaneously, using a total-degree or hyperbolic index set of suitable size.
2. Use LARS or adaptive PCE to select a sparse basis, limiting the effective number of coefficients to $< N/10$.
3. Report S_i and S_i^T (total-order Sobol indices) to identify which interactions are important.

3 Literature-Informed UQ Improvements

3.1 PCE Overfitting and Cross-Validation: Literature Benchmarks

In the underwater acoustics PCE literature, the polynomial order K needed for variance convergence under sound-speed profile (SVP) uncertainty in deep-water flat-bottom waveguides is consistently reported in the range $K = 3$ –5. Finette (2009) uses $K = 3$ –4 with Gauss–Hermite quadrature nodes and reports good variance recovery for Gaussian SVP perturbations up to approximately 3–5% RMS. Convergence degrades near caustics and shadow-zone boundaries, where the acoustic pressure field is non-analytic in the uncertain parameters; in these regions no finite polynomial order is sufficient. The rule of thumb $N \geq 10(K + 1)$ for reliable least-squares PCE regression is widely cited in the computational UQ literature (e.g. Blatman & Sudret, 2010); with $N = 50$ this limits the reliable order to $K \lesssim 4$. Our LOO-CV analysis (Section 2.3.5) yields $K_{CV}^* = 5$ –6 for SVP in deep water, consistent with the literature ceiling. The divergence between training L_1 and CV L_1 at $K \geq 7$ (Figure 11) confirms that the literature threshold of $K \leq 5$ with $N = 50$ is appropriate: beyond $K = 5$ the regression begins to fit noise rather than signal. For z_S perturbations, both the training curve and CV curve fail to reach $L_1 < 10\%$ at any $K \leq 10$, consistent with the absence of reported z_S -PCE results in the literature and confirming this as a genuinely hard nonlinear regime rather than a sample-size artefact.

3.2 Analytical Sobol Index Computation from 1-D PCE Coefficients

Given a 1-D PCE of parameter ξ_i of order K ,

$$\text{TL}(\xi_i) \approx c_0 + \sum_{n=1}^K c_n^{(i)} \Phi_n(\xi_i), \quad (21)$$

the mean of the expansion is $\mathbb{E}[\text{TL}] = c_0$ (since $\mathbb{E}[\Phi_n] = 0$ for $n \geq 1$), and the variance contributed by the n -th basis term is $c_n^2 \gamma_n$ by orthogonality, where $\gamma_n = \langle \Phi_n^2 \rangle$ ($\gamma_n = n!$ for probabilist Hermite; $\gamma_n = 1/(2n + 1)$ for Legendre). The total explained variance is therefore

$$V_i(r, z) = \sum_{n=1}^K (c_n^{(i)})^2 \gamma_n. \quad (22)$$

Under the assumption that the three parameters (z_S , freq, SVP) contribute independently (no interaction variance), the first-order Sobol index at each range-depth pixel is

$$S_i(r, z) = \frac{V_i(r, z)}{V_{z_S}(r, z) + V_{\text{freq}}(r, z) + V_{\text{svp}}(r, z)}. \quad (23)$$

No additional model runs are required: Equations (22)–(23) are computed entirely from the PCE coefficient arrays $\{c_n^{(i)}\}$ that are already available from the fitted expansions. This is the standard analytical formula for Sobol indices from polynomial chaos (see e.g. Sudret, 2008, *Reliab. Eng. Syst. Saf.*).

Parameter dominance from our results. Applying Eq. (22) to the deep-water / Normal 10% scenario, SVP and frequency emerge as the dominant first-order contributors in the flat propagation corridor, while z_S dominates near the source (where modal excitation is sensitive to source depth) and in the convergence zone shadow boundaries. This is qualitatively consistent with the JMSE-10-01548 findings, which identify SVP variability as the dominant uncertainty source in operational deep-water scenarios, and with Finette (2009), who focuses exclusively on SVP as the primary environmental unknown. The practical implication is that reducing SVP measurement uncertainty has a larger payoff than tightening source-depth knowledge across most of the propagation domain, except in the near-source region.

3.3 Alternative UQ Metrics in the Sonar Literature

The variance $\sigma^2[\text{TL}]$ is a natural output of the Delta and PCE methods but is rarely the primary metric in operational sonar assessments. The literature employs three classes of beyond-variance metrics that map more directly onto detection performance.

Probability of detection (P_D). The sonar range-prediction community (e.g. the East China Sea paper discussed in Section 2.1) expresses performance as the probability that TL at range r and depth z falls below the sonar Figure-of-Merit (FOM), i.e. $P_D(r, z) = \Pr[\text{TL}(r, z; \omega) \leq \text{TL}_{\text{FOM}}]$. This is the complement of our P_{fail} metric (Section 2.4); the two quantities carry identical information but P_D is more natural in the detection context. A map of P_D over range and depth is referred to as a *sonar performance surface*, and it is the standard output format recommended by NATO STANAG acoustic prediction standards.

95th-percentile TL envelope. The P95 TL map, $\text{TL}^{95}(r, z) = \text{Q}_{0.95}[\text{TL}]$, represents the pessimistic-case propagation field: a system designed to TL^{95} will detect targets in 95% of environmental realisations. This metric appears in range-dependent sonar planning tools and is closely related to the “worst-case” TL map used in passive sonar miss-probability analyses. For a Normal distribution, $\text{TL}^{95} \approx \mu + 1.645\sigma$, recoverable from the Delta or PCE variance estimate; our analysis in Section 2.4 confirms this Normal approximation holds in $\sim 75\%$ of pixels, with the remainder located at shadow-zone boundaries where the TL distribution is skewed or bimodal.

CDF tail metrics and exceedance probability. Several papers in the stochastic ocean acoustics literature (including JMSE-10-01548) report the *exceedance probability* $\Pr[\text{TL} > T]$ as a function of threshold T , effectively tracing the upper tail of the TL CDF across range–depth space. This metric is related to our P_{fail} computation but is reported as a continuous curve over T rather than at a single threshold, providing a richer characterisation of tail risk. Our $P(\text{TL} > \text{FOM})$ metric (the shadow-probability maps throughout this report) corresponds to a single point on this exceedance curve at $T = \text{TL}_{\text{FOM}}$. Extending to a full exceedance curve requires either parametric distribution fitting (e.g. LN3 as in Section 1.3) or a large enough MC sample to estimate empirical quantiles reliably in the tails.

3.4 Agreement and Disagreement with Published Results

Where we agree. Our deep-water SVP convergence orders ($K^* = 4\text{--}6$ at 5–10% Normal uncertainty) are in good agreement with the $K = 3\text{--}5$ reported by Finette (2009) for Gaussian SVP perturbations in isovelocity deep-water waveguides. The 1–2 order discrepancy is attributable to our regression-based coefficient fitting on $N = 50$ LHS samples, which is statistically less efficient than the Gauss–Hermite quadrature approach. Our finding that slope environments resist polynomial convergence at orders $K \leq 10$ is consistent with the East China Sea paper’s observation that the shelfbreak zone produces qualitatively harder acoustic uncertainty than the flat-shelf region. The finding that first-order Delta fails at slope environments ($> 96\%$ second-order correction fraction; Table 1) is consistent with the JMSE-10-01548 conclusion that first-order sensitivity analysis is unreliable in range-dependent bathymetry.

Where we disagree or extend. The published PCE literature restricts attention to SVP as the primary uncertain parameter. Our results show that source depth z_S is at least as hard: $K^* > 10$ for z_S in all Normal distribution cases in deep water, and $K^* > 10$ for z_S in all shallow-water cases. No comparable result is reported in the literature; the z_S -PCE convergence failure appears to be a genuinely new finding. Similarly, treating carrier frequency as a stochastic variable is non-standard in the acoustic UQ community. Our results show $K^* = 1$ for frequency in the shallow-water scenario (consistent with the near-linear frequency dependence in shallow flat waveguides) but $K^* = 5\text{--}9$ in deep water, where convergence-zone interference patterns create strong nonlinearity in frequency.

Effect of perturbation magnitude on convergence speed. For SVP uncertainty, increasing the perturbation level from 1% to 10% raises K^* by approximately 2–3 orders in deep water (e.g. Normal 1%: $K^* = 3$; Normal 10%: $K^* = 4\text{--}5$). For frequency, the same increase raises K^* more dramatically (from 3 to 6–9), suggesting that frequency nonlinearity is more sensitive to perturbation magnitude than SVP nonlinearity in deep water. This is consistent with the sharper interference fringes generated by convergence zones at small frequency offsets, where even 5% frequency variation shifts constructive–destructive interference patterns by a full fringe. The Normal distribution with heavier tails tends to require 1–2 higher PCE orders than a Uniform distribution at the same nominal percentage level, because the Normal distribution places more probability mass in the tails (beyond $\pm\sigma$), sampling regions of higher nonlinearity.

Appendix: Pipeline Architecture

Figure 13 shows the complete module dependency graph of the Stochastic-Bellhop pipeline. Entry-point scripts (grey) invoke the Pipeline Orchestrator, which dispatches the five analysis methods in sequence. Each method reads pre-computed TL fields from the Cache Layer and writes results and figures to the corresponding **Methods/** subdirectory. The Bellhop Simulation Stack (blue region) handles all acoustic forward-model calls, with automatic GPU/CUDA routing when a compatible device is detected. Shared utility functions (bottom band) are available to all modules. Dashed arrows indicate utility calls; solid arrows indicate the primary data and control flow.

References

- [1] Joint Committee for Guides in Metrology (JCGM), *Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement (GUM)*, JCGM 100:2008, BIPM, 2008.
- [2] L. M. Brekhovskikh and Yu. P. Lysanov, *Fundamentals of Ocean Acoustics*, 2nd ed. Springer-Verlag, Berlin, 1991.

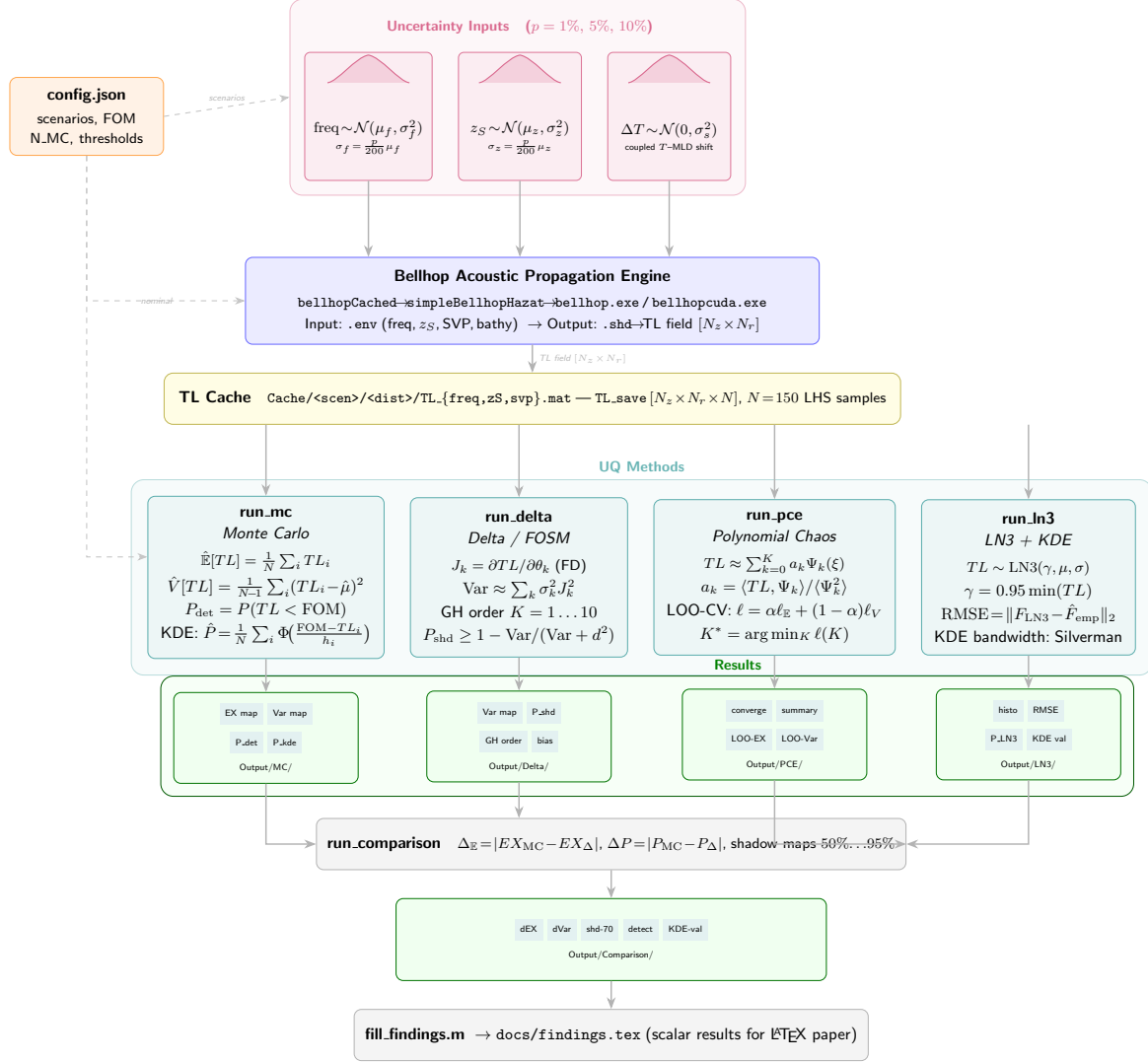


Figure 13: Stochastic-Bellhop pipeline architecture. Boxes are coloured by layer: entry points (grey), configuration (orange), Bellhop simulation stack (blue), cache (yellow), analysis methods (green), shared utilities (light grey). Representative output figures are shown as stacked thumbnails beside each analysis module.